

Propagate by seed sown under glass in spring.

C. argentea. Moderately fast-growing, erect, bushy perennial, usually grown as an annual. H 12–24in (30–60cm), S to 18in (45cm). Has ovate to lance-shaped, pale to mid-green leaves and, in summer, silvery-white, pyramid-shaped, feathery flower heads, to 4in (10cm) long. Cultivars are available in red, orange, yellow, and cream. Z10–11. **‘Armor Mixed’**, H 12–16in (30–40cm), S 4–6in (10–15cm), has crested, curled heads of tightly packed, orange, purple, red, or yellow flowers.

Century Series cultivars are among the most vigorous and widely cultivated, bearing vivid red, rose-pink, or yellow flower heads. **var. cristata ‘Dragon’s Breath’**, H 2ft (60cm), S 16in (40cm), has purple-tinged leaves and long-lasting, upright, flame-red flower heads. **var. cristata Kimono Series** cultivars, H to 8in (20cm), are dwarf, bearing large flower heads in bright colors, including salmon-pink, rose-red, yellow, or creamy-white.

CELTIS

Hackberry, Nettle tree

CANNABACEAE

Genus of deciduous and evergreen trees and shrubs, with inconspicuous flowers in spring, grown for their foliage and small fruits. Needs full sun (best in hot summers) and fertile, well-drained soil. Propagate by seed in fall.

C. australis illus. p.62.

C. occidentalis. Deciduous, spreading tree. H and S 70ft (20m). Ovate, sharply toothed, glossy, bright green leaves turn yellow in fall, when they are accompanied by round, yellowish-red, then red-purple fruits. Z2–9

C. sinensis. Deciduous, rounded tree. H and S 30ft (10m). Has ovate, glossy, dark green leaves, with fine teeth, and small, round, orange fruits. Z2–9

CENTAUREA

Knapweed

ASTERACEAE/COMPOSITAE

Genus of annuals, biennials, perennials and subshrubs, grown for their flower heads that each have a thistlelike center surrounded by a ring of slender ray petals. Requires sun; grows in any well-drained soil, even poor soil. Propagate by seed or by division in fall or spring.

C. cyanus (Cornflower). Fast-growing, upright, branching annual. H 1–3ft (30cm–1m), S 1ft (30cm). Has lance-shaped, gray-green leaves and, in summer and early fall, branching stems with usually double, daisylike, dark blue flower heads. Flowers are excellent for cutting. Tall (blue, illus. p.315) and dwarf cultivars are available, with flowers in shades of blue, pink, red, purple, or white. Z2–11.

Baby Series (dwarf), H to 1ft (30cm), has blue, white, or pink flower heads.

C. dealbata. Erect perennial. H 3ft (1m), S 2ft (60cm). Lilac-purple flower heads are borne freely in summer, one or more to each stem. Has narrowly ovate, finely cut, light green leaves. Z2–11. **‘Steenbergii’**, H 2ft (60cm), has carmine-lilac flowers.

C. hypoleuca ‘John Coutts’ illus. p.265.

C. macrocephala. Robust, clump-forming perennial. H 3ft (1m), S 2ft (60cm). In summer, sturdy stems bear large, yellow flower heads, within papery, silvery-brown bracts. Mid-green leaves are narrowly ovate and deeply cut. Z2–9

C. montana illus. p.269.

C. moschata. See *Amberboa moschata*.

C. pulcherrima illus. p.232.

CENTRANTHUS

Valerian

CAPRIFOLIACEAE

Genus of late spring- to fall-flowering annuals and perennials, a few subshrubby. Requires sun. Thrives in an exposed position and in poor, alkaline soil. Propagate by seed in fall or spring.

C. ruber (Red valerian) illus. p.248.

CENTROPOGON

CAMPANULACEAE

Genus of herbaceous or subshrubby, upright, scrambling or climbing, deciduous or evergreen perennials, grown for their tubular, bright pink to red, orange or yellow flowers borne singly on stalks from between the uppermost leaves, which are sometimes patterned or variegated. Needs humus-rich, moist, but well-drained soil in sun or partial shade. Propagate by seed in spring.

C. ayavacensis, syn. *C. willdenowianus*. Deciduous, trailing perennial. H and S 3ft (1m). Flexuous, purple stems have rounded to broadly elliptic, mid-green leaves. Bears tubular, cerise flowers, with reflexed petals, from early spring to summer. Needs a sunny position. Protect roots from frost. Z11–12 (min. 50°F/10°C)

C. cordifolius. Herbaceous, weakly climbing or trailing perennial. H 20in (50cm), S 3–6ft (1–2m). Red stems have heart-shaped, toothed, gray-green leaves, and bear flared, tubular, cerise flowers, from late spring to summer. Requires a sunny position. Protect roots from frost. Z11–12 (min. 50°F/10°C)

C. ferrugineus, syn. *Lobelia ferruginea*. Herbaceous, trailing perennial. H ½–6ft (0.5–2m), S 3–6ft (1–2m). Long, wiry, pendent stems bear narrowly ovate, silver-marked, mid-green leaves, with narrowly toothed margins. Flared, tubular, bright pink flowers, from summer into fall, are followed by round fruits. Best in a container. Z11–12 (min. 50°F/10°C)

C. willdenowianus. See *C. ayavacensis*.

CEPHALARIA

CAPRIFOLIACEAE

Genus of annuals and perennials, best suited to large borders and wild gardens. Prefers sun and well-drained soil. Propagate by division in spring or by seed in fall.

C. gigantea, syn. *C. tatarica*, *Scabiosa gigantea*, *S. tatarica*, (Giant scabious, Yellow scabious). Robust, branching perennial. H 6ft (2m), S 4ft (1.2m). In early summer, wiry stems bear pincushionlike heads of primrose-yellow flowers above lance-shaped, deeply cut, dark green leaves. Z4–7

C. tatarica. See *C. gigantea*.

CEPHALOCEREUS

CACTACEAE

Genus of slow-growing, columnar, perennial cacti with 20–30-ribbed, green stems. Prefers full sun and extremely well-drained, lime-rich soil. Prone to rot if overwatered. Propagate by seed in spring or summer.

C. senilis, syn. *Pilocereus senilis* (Old-man cactus), illus. p.479.

CEPHALOPHYLLUM

AIZOACEAE

Genus of creeping, clump-forming or spreading, perennial succulents with semicylindrical to cylindrical, green leaves. Flowers are borne after 1 or 2 years. Needs sun and well-drained soil. Propagate by seed in spring or summer.

C. alstonii. Prostrate, perennial succulent. H 4in (10cm), S indefinite. Has cylindrical, gray-green leaves, to 3in (7cm) long. Bears daisylike, dark red flowers, 3in (8cm) across, in summer. Z10–12 (min. 45°F/7°C)

C. pillansii. Clump-forming, perennial succulent. H 3in (8cm), S 2ft (60cm). Leaves are cylindrical, 2½in (6cm) long, dark green and covered in darker dots. Short flower stems produce daisylike, red-centered, yellow flowers, 2½in (6cm) across, from spring to fall. Z10–12 (min. 45°F/7°C)

CEPHALOTAXUS

Plum yew

TAXACEAE/CEPHALOTAXACEAE

See also CONIFERS.

C. harringtonii (Cowtail pine, Plum yew). Bushy, spreading conifer. H 15ft (5m), S 10ft (3m). Needlelike, flattened leaves are glossy, dark green, grayish beneath, radiating around erect shoots. Bears egg-shaped, fleshy, green fruits that ripen to brown. Z6–9. **‘Fastigiata’**, H and S 15ft (5m), is shrubby, with erect branches and radially arranged leaves, to 3in (8cm) long.

CERASTIUM

CARYOPHYLLACEAE

Genus of annuals and perennials with star-shaped flowers. Some species are useful as groundcover. Needs sun and well-drained soil. Propagate by division in spring.

C. alpinum (Alpine mouse-ear). Prostrate perennial. H 3in (8cm), S 16in (40cm). Tiny, ovate, gray leaves cover stems. Flower stems carry solitary, small, star-shaped, white flowers throughout summer. Z2–9

C. tomentosum (Snow-in-summer) illus. p.350.

CERATOPHYLLUM

Hornwort

CERATOPHYLLACEAE

Genus of deciduous, perennial, submerged water plants, grown for their foliage. Suitable for ponds and cold-water aquariums. Prefers an open, sunny position, but tolerates shade better than most submerged plants. Propagation occurs naturally when scaly young shoots

or winter buds separate from main plants. Take stem cuttings in growing season.

C. demersum (Hornwort). Deciduous, perennial, spreading, submerged water plant that occasionally floats. S indefinite. Has small, dark green leaves with 3 linear lobes. Best suited to a cool-water pond. Z5–11

CERATOPTERIS

PTERIDACEAE

Genus of deciduous or semievergreen, perennial or annual, submerged or floating water ferns, grown for their attractive foliage. Suitable for aquariums. Prefers a sunny position. Remove fading fronds regularly. Propagate in summer by division or by buds that develop on the leaves.

C. thalictroides (Water fern). Semievergreen, perennial, spreading, floating water fern that sometimes roots and becomes submerged. S indefinite. Lance-or heart-shaped, soft green fronds are wavy-edged. Z11–12 (min. 50°F/10°C)

CERATOSTIGMA

PLUMBAGINACEAE

Genus of deciduous, semievergreen or evergreen subshrubs and herbaceous perennials, grown for their blue flowers and fall color. Requires a sunny position, with well-drained soil. Cut out old, dead wood from shrubs in spring. Propagate shrubs by softwood cuttings in summer, perennials by division in spring.

C. griffithii. Evergreen or semievergreen, bushy, dense shrub. H 3ft (1m), S 5ft (1.5m). Spoon-shaped, bristly, purple-edged, dull green leaves redden in fall. Clusters of tubular, bright blue flowers, with spreading petal lobes, are borne in late summer and fall. Z7–10

C. plumbaginoides illus. p.346.

C. willmottianum illus. p.159. **FOREST BLUE (‘Lice’)** is a compact, bushy, deciduous shrub. H 3ft (1m), S 5ft (1.5m). Lance-shaped to ovate, pointed, mid- to dark green, purple-margined leaves, turn rich red in fall. Bears abundant, tubular, cobalt-blue flowers, with spreading petal lobes, in late summer and fall. Z7–10

CERCIDIPHYLLUM

CERCIDIPHYLLACEAE

Genus of deciduous trees, grown for their foliage and often spectacular fall color. Late frosts may damage young foliage, but do not usually cause lasting harm. Requires sun or semishade and fertile, moist, but well-drained soil. Propagate by seed in fall.

C. japonicum (Katsura) illus. p.66.

CERCIS

Judas tree, Redbud

CAESALPINIACEAE

Genus of deciduous shrubs and trees with sometimes shrubby growth, cultivated for their foliage and small, pealike flowers, borne profusely in spring. Requires a position in full sun with deep, fertile, well-drained soil. Plant out as young specimens. Resents